

Statistical Analysis Plan (SAP)

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Title	Relative Effectiveness of Social Media, Dating Apps, and Information Search Sites in Promoting HIV Self-testing: Observational Cohort Study
CRU/Department/Division/Center	
IRB Number	#18-001580
Investigators:	Jeffrey D. Klausner, M.D.
Primary Investigator	
Collaborative Lead	
Co-authors (if known)	Chrysovalantis Stafylis, Gabriella Vavala, Qiao Wang, Bethany McLeman, Shea M Lemley, Sean D Young, Haiyi Xie, Abigail G Matthews, Neal Oden, Leslie Revoredo, Dikla Shmueli-Blumberg, Emily G Hichborn, Erin McKelle, Landhing M Moran, Petra Jacobs, Lisa A Marsch
Analysis Biostatistician(s)	
Biostatistics Supervisor	
Lead Biostatistician	Haiyi Xie
Subject Matter Expert	
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Project Goal(s)	Manuscript
Submission Deadline(s)	2022 Sep 23
Effort Estimate (optional)	

Investigator Agreement	☐ All statistical analyses included in an abstract or manuscript should reflect the work of the biostatistician(s) listed on this SAP. No changes or additional analyses should be made to the results or findings without discussing with the project biostatistician(s).
	☐ All biostatisticians on this SAP should be given sufficient time to review the full presentation, abstract, manuscript, or grant and be included as co-authors on any abstract or manuscript resulting from the analyses.
	☐ If substantial additional analysis is necessary or the aims of the project change, a new SAP will need to be developed.
	☐ Publications resulting from this SAP are supported in part by the Duke CTSA and must cite grant number UL1TR002553 and be submitted to PubMed Central.
	☐ I have reviewed the SAP and understand that any changes must be documented.

Acknowledged by:
Date:

Activity Log

SAP Version	Date	Change(s)	Rationale
Version 1.0		Original statistical analysis plan	

	Version 2.0		Update to statistical analysis plan	Due to significant wave-platform interactions, instead of pooling sites of the same platform type to compare test kit order rates across platform types, the sites per platform were separated and compared within waves (pairwise comparison with Hochberg adjustment on p-value).
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Acronyms	MSM	Men who have sex with men
	PrEP	pre-exposure prophylaxis
	HIV	human immunodeficiency virus
	AIDS	acquired immunodeficiency syndrome

1 Study Overview

Background/Introduction:

HIV is a sexually transmitted infection that attacks the human immune system, and results in AIDS at the most advanced stage of infection. Once they get HIV, a patient cannot be cured. Therefore, prevention and treatment efforts for HIV are very important. In the 1980s-1990s, the HIV pandemic swept the globe, and in particular devastated queer communities due to many gay men passing away from the disease. Men who have sex with other men, referred to as MSM in the study, are at particular risk of HIV if they have anal sex without protection. PrEP, the prevention method discussed in this study which is meant to be taken as a daily pill, has been shown to be very effective at reducing the risk of infection. Young MSM from ages 18-24, who are part of minority groups (Black/Latinx) are a group that are at risk but do not often know of or use PrEP. They also may engage in other behaviors that could put them at higher risk of HIV, such as binge drinking or drug usage. Because of this, spreading awareness for HIV prevention and examining its relationship to other risk factors is particularly important.

1.1 Study Aims

Primary aim: To determine the difference in effectiveness between several different forms of internet outreach, in particular social media platforms, dating apps, and search engines, in encouraging MSM who are at higher risk of HIV to engage in HIV prevention efforts. This is determined by the number of HIV test kits that they order.

Secondary aims:

Examined if there was a correlation between the primary outcome (number of self-test kits ordered) and the following variables:

- 1) Reported substance use
- 2) Stage of health behavior change for HIV testing (Precontemplation, Contemplation, Determination, Action, Maintenance)
- 3) Attitudes toward HIV testing and treatment
- 4) HIV-related stigma
- 5) Medical mistrust
- 6) Opinions about PrEP measures

Exploratory aim:

According to the manuscript, "As an exploratory outcome, we recorded the advertisement metrics of each campaign to measure differences in the reach and cost."

1.2 Study Hypotheses

1.2.1 Primary Hypotheses

Let r_{ij} denote test kit order rate for wave number i and platform number j . The number of test kits ordered per platform is assumed to follow a Poisson process with this rate.

Where $i \in \{1, 2\}$ and $j \in \{1, 2, 3\}$ for {social media, dating apps, search engine}

Null hypothesis: $r_{11} = \dots = r_{23}$

Alternative hypothesis: $r_{ij} \neq r_{i'j'}$ for some i, j, i', j' where $i \neq i'$ and $j \neq j'$

1.2.2 Secondary Hypotheses

Null hypothesis 1: There is no association between reported substance use and the rate of test kit ordering.

Alternative hypothesis 1: There is some association between reported substance use and the rate of test kit ordering.

Null hypothesis 2: There is no association between the stage of health behavior change for HIV testing and the rate of test kit ordering.

Alternative hypothesis 2: There is some association between stage of health behavior change for HIV testing and the rate of test kit ordering.

Null hypothesis 3: There is no association between attitudes toward HIV testing and treatment and the rate of test kit ordering.

Alternative hypothesis 3: There is some association between attitudes toward HIV testing and treatment and the rate of test kit ordering.

Null hypothesis 4: There is no association between HIV-related stigma and the rate of test kit ordering.

Alternative hypothesis 4: There is some association between HIV-related stigma and the rate of test kit ordering.

Null hypothesis 5: There is no association between medical mistrust and the rate of test kit ordering.

Alternative hypothesis 5: There is some association between medical mistrust and the rate of test kit ordering.

Null hypothesis 6: There is no association between opinions about PrEP measures and the rate of test kit ordering.

Alternative hypothesis 6: There is some association between opinions about PrEP measures and the rate of test kit ordering.

2 Study Population

2.1 Inclusion Criteria

As stated in the protocol, the participant was included if they had all the following traits:

- Have clicked on one of the study-specific advertisements posted on the platforms/ websites;
- Have been biologically born male (cis-gender man), per participant self-report;
- Report condomless anal intercourse AND more than one male sex partner in the 90 days prior to the date of the screening questionnaire;
- Be between the ages of 18-30 years old, inclusive;
- Self-identify as Latino and/or Black/African American; (multiracial is also OK)
- Not currently on PrEP and haven't taken PrEP in the last six months prior to the date of the screening questionnaire (per participant self-report);
- Have not tested for HIV in the last 3 months prior to the date of the screening questionnaire (per participant self-report);
- Have a Facebook account (for identity validation to reduce duplicate attempts at enrollment);
- Be willing to provide contact information (phone number, email) to the study team.

2.2 Exclusion Criteria

As stated in the protocol, the participant was excluded if they met any of the exclusion criteria:

- Were HIV-positive already, less than 30 days prior to the date of screening
- Were currently undergoing treatment for HIV infection
- Unwilling or unable to provide informed consent.
- Unwilling to provide contact information (phone number, email address).

2.3 Data Acquisition

Fill in all relevant information:

Study design	Longitudinal, observational study, non-randomized
Data source/how the data were collected	Online survey
Contact information for team member responsible for data collection/acquisition	Unknown
Date or version (if downloaded, provide date)	Unknown
Data transfer method and date	Unknown
Where dataset is stored	Unknown

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

The primary outcome of this study is the number or rate of HIV home self-test kits ordered per day for each internet platform used. The variable name is ORA_REDEEMED, where there are 3 categories per study participant: “Yes” if a self-test kit was redeemed, “No” if a self-test was not redeemed, and “Over 60 Days” if it took over 60 days to redeem (which is counted as “No”).

3.2 Secondary Outcome(s)

The secondary outcomes for this study are related to investigating if other factors such as participants’ substance use affect the number of test kits ordered.

Outcome	Description	Variables and Source	Specifications
Reported substance use	Degree of substance use per participant.	Q13_1,Q13_2,Q13_3,Q13_4	For each question relating to each substance, the usage is coded as 1 if the participant agrees to the question and 0 if they disagree. For each substance, take the sum across all questions asking about that substance, and then assign as follows: 0 total = “None” 1 total = “Problem use” ≥ 2 total = “High risk use” For stimulants, opioids, sedatives and prescribed stimulants, the problem use and high risk use categories are combined into one due to there not being many people in those two categories.
Stage of health behavior change for HIV testing	Gauging how willing the participant’s attitude is towards HIV testing.	Q15_1	1 = Precontemplation (I don't see any need to regularly test for HIV) 2 = Contemplation (I think I should get tested for HIV regularly, but I am not sure) 3 = Determination (I'm ready to start getting regularly tested for HIV) 4 = Action (I'm trying to get tested regularly for HIV) 5= Maintenance (I've been getting testing for HIV regularly over the past few years.)

Attitudes toward HIV testing and treatment	Several aspects of the participant's attitude towards HIV testing and treatment.		The questions are related to agreeing(1) or disagreeing (2) for the following statements:
		Q15_3	Getting tested for HIV helps people feel better.
		Q15_4,	Getting tested for HIV helps people from getting HIV.
		Q15_5,	People in my life would leave if I had HIV.
		Q15_6,	People who tested positive for HIV should hide it from others.
		Q15_7	I would rather not know if I have HIV.
HIV-related stigma	Participant opinions on statements relating to stigma against those with HIV.		Each question is coded as one of the numbers 1 - 7 (Strongly Disagree, Disagree, Somewhat Disagree, Neither agree nor disagree, Somewhat Agree, Agree, Strongly Agree) for the participant's opinion on the following statements:
		Q14_2	I feel afraid of people living with HIV/AIDS.
		Q14_3	I could not be friends with someone who has HIV/AIDS.
		Q14_4	People who get HIV/AIDS through sex or drug use got what they deserve.
		Q14_5	I feel anger toward people with HIV/AIDS.
Medical mistrust	Participant's response to several questions that determine how much trust they have in medical organizations.		For Q16_1, the coding is 28=Strongly agree; 30=Agree; 33=Disagree; 34=Strongly disagree with the statement, while for the other 6 questions, the coding is 1=Strongly agree; 2=Agree; 6=Disagree; 7=Strongly disagree Corresponding statements:
		Q16_1	You'd better be cautious when dealing with healthcare organizations.

		Q16_2	Patients have sometimes been deceived or misled by health care organizations.
		Q16_3	When health care organizations make mistakes they usually cover it up.
		Q16_4	Health care organizations have sometimes done harmful experiments on patients without their knowledge.
		Q16_5	Health care organizations don't always keep your information totally private.
		Q16_6	Sometimes I wonder if health care organizations really know what they are doing.
		Q16_7	Mistakes are common in health care organizations.
Opinions about PrEP measures	Participant opinion on HIV treatment.		Each question is a continuous sliding scale from 1=Strongly Disagree to 7=Strongly Agree for the participant's opinion on the following statements:
		Q94_1	I am less threatened by the idea of being HIV positive than I used to be.
		Q94_5	I am less worried about HIV infection than I used to be
		Q94_6	I think HIV/AIDS is less of a problem than it used to be
		Q94_7	I think HIV/AIDS is a less serious threat than it used to be because of new HIV/AIDS treatments
		Q94_8	I am much less concerned about becoming HIV positive myself because of new HIV/AIDS treatments
		Q94_9	I think that condom use during sex is less necessary now that new HIV/AIDS treatments are available
		Q94_10	I think that someone who is HIV positive now needs to care less about condom use

		Q94_11	I think that the need for condom use is less than it used to be, because you can always start new treatments
		Q94_12	I think that someone who is HIV positive and uses new HIV/AIDS treatments can be cured
		Q94_13	I think that new HIV/AIDS treatments can eradicate the virus from your body

3.3 Additional Variables

Outcome	Description	Variables and Source	Specifications
Flag for participant inclusion in the final analysis		PO_FLAG	1 for included, 0 for not
Platform type		SITE_TYPE	Possible values: "Social Media", "Info sites", "Dating App"
Name of site		SITE	Possible values: "Facebook", "Instagram", "Google", "Bing", "Grindr", "Jack'd"
Wave number		WAVE	1, 2, or 4 (4 is combined with 1 for final analysis)
Time the wave was active			Not present in the original dataset, but 38 days for Wave 1 and 70 days for Wave 2.
Race		Q5_3	25= American Indian or Alaska Native; 26= Asian; 24= Black or African American; 27= Native Hawaiian or Pacific Islander; 23= White; 28= Other, please specify
Age		Q3_1	Numeric
Ethnicity	Are you Hispanic/Latino?	Q5_1	1=Yes; 2=No
History of PrEP uptake	Have you ever taken PrEP?	Q6_2	1 = Yes, but I haven't taken PrEP in the past 6 Months; 2= Yes, but I am currently taking PrEP; 3=No, I have never taken PrEP

Condom use	How often do you use condoms?	Q11_3	1 = "Never" 2= "Sometimes" 3= "About half the time", 4 = "Most of the time", 5= "Always"
Number of sexual partners	How many male sexual partners have you found in 90 days?	Q11_2	Whole numbers
HIV testing history	Have you ever been tested for HIV in your lifetime?	Q11_5	1=Yes, 2=No
Reasons to not get tested for HIV	If you have not been tested for HIV, which one of the following would you say is the MAIN reason why you have not been tested?	Q11_7	1= It's unlikely you've been exposed to HIV; 2 = You are afraid to find out if you are HIV positive; 3 = You didn't want to think about HIV or about being HIV positive; 4 = You were worried your name would be reported to the government if you tested positive; 5 = You don't like needles; 6 = You don't trust the results to be confidential; 7 = You didn't know where to get tested; 8= Some other reason. Please specify.

4 Statistical Analysis Plan

This is a longitudinal observational study that is not randomized. After clicking on the ad on one of the online platforms, participants are shown information on the study and can opt-in to participate. After completing the baseline evaluation, they are told to order an HIV self-test kit, and then there are 2 follow-up surveys at 14 and 60 days. For the 14-day follow-up, the participants are asked about their usage of PrEP and if they sought medical care if they tested positive for HIV. For the 60-day follow-up, they are then asked about some of the secondary outcomes like their attitudes around PrEP. Thus they are in the study for around 2 months.

4.1 Demographic and Clinical Characteristics ("Table 1")

The variables in Table 1 are as follows:

Outcome	Description	Variables and Source	Specifications
Race		Q5_3	25= American Indian or Alaska Native; 26= Asian; 24= Black or African American; 27= Native Hawaiian or Pacific Islander; 23= White; 28= Other, please specify
Age		Q3_1	Whole numbers
Ethnicity	Are you Hispanic/Latino?	Q5_1	1=Yes; 2=No
History of PrEP uptake	Have you ever taken PrEP?	Q6_2	1 = Yes, but I haven't taken PrEP in the past 6 Months; 2= Yes, but I am currently taking PrEP; 3=No, I have never taken PrEP
Condom use	How often do you use condoms?	Q11_3	1 = "Never" 2= "Sometimes" 3= "About half the time", 4 = "Most of the time", 5="Always"
Number of sexual partners	How many male sexual partners have you found in 90 days?	Q11_2	Whole numbers
HIV testing history	Have you ever been tested for HIV in your lifetime?	Q11_5	1=Yes, 2=No
Reasons to not get tested for HIV	If you have not been tested for HIV, which one of the following would you say is the MAIN reason why you have not been tested?	Q11_7	1= It's unlikely you've been exposed to HIV; 2 = You are afraid to find out if you are HIV positive; 3 = You didn't want to think about HIV or about being HIV positive; 4 = You were worried your name would be reported to the government if

			you tested positive; 5 = You don't like needles; 6 = You don't trust the results to be confidential; 7 = You didn't know where to get tested; 8= Some other reason. Please specify.
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4.2 Analyses Plan for Aim 1

For the primary outcome, a Poisson regression model will be used to model the response (number of test kits ordered) with a log link. According to Appendix 1, with wave number i and platform number j , the outcome o_{ij} is given by:

$$\log(o_{ij}) = \log(t_i) + \alpha + \beta_i + \gamma_j + \beta\gamma_{ij}$$

Where:

t_i is the time that the wave i platforms were recruiting.

α is the intercept/grand mean.

β_i is the main effect of wave i .

γ_j is the main effect of platform type j .

$\beta\gamma_{ij}$ is the interaction term between wave and platform effect.

The corresponding rate for the Poisson process is given by:

$$r_{ij} = \exp(\alpha + \beta_i + \gamma_j + \beta\gamma_{ij})$$

These rates will be estimated from the data as well as their 95% confidence intervals.

The original plan was to compare rates within waves across platforms:

	Social Media	Dating Apps	Info Sites
Wave 1	Facebook	Grindr	Google
Wave 2	Instagram	Jack'd	Bing

- Social Media vs Dating Apps
- Social Media vs Info Sites
- Dating Apps vs Info Sites

But due to the fact that there was not enough recruitment in Wave 1, each platform was compared to every other platform in the same wave individually:

- Facebook vs Grindr
- Facebook vs Google
- Grindr vs Google
- Instagram vs Jack'd
- Instagram vs Bing
- Jack'd vs Bing

Additionally, there was originally a third wave (Twitter, Yahoo, and Hornet), but this was excluded from the final analysis due to no participants signing up and no test kits being ordered.

4.3 Analyses Plan for Aim 2

For the secondary and exploratory outcomes, univariate tests and 95% confidence intervals will also be used.

For each of the six secondary aims, the participants are split into 2 groups: those who did order a self-test kit within 60 days, and those who did not. Then, their responses to each of the secondary aims are summed per category, and a statistical test on the resulting contingency table is ran to determine if the 2 groups were significantly different in their responses.

- Reported substance use: Fisher's exact test
- Stage of health behavior change for HIV testing: Fisher's exact test
- Attitudes toward HIV testing and treatment: Wilcoxon rank test
- HIV-related stigma: Fisher's exact test
- Medical mistrust: Wilcoxon rank test
- Opinions about PrEP measures: Wilcoxon rank test

4.4 Analyses Plan for Aim 3

For the exploratory aim, the plan is to record the money spent per platform, and the clicks, impressions, and click through rates of each.

5 Limitations

There is an inherent bias to this study design, as it depends on survey responses, which are entirely voluntary, meaning that it is more likely for people with more extreme experiences to bother responding. People may not choose to opt-in to the study at all after clicking on the ad to learn more about PrEP/HIV prevention.

Additionally, the focus on rate of HIV test kit ordering doesn't take into account the differences in number of people who use each type of internet platform.

As according to the manuscript, the study was done in states that were known for having high rates of HIV, so it may not be generalizable to the rest of the US. Additionally, there was low engagement with some of the waves, for example with Bing having zero participants, which impacted the quality of the analysis. Finally, these findings are specific to the sites selected, as they were selected due to their popularity and grouped into platforms based on similarities (such as Facebook and Instagram both counting as social media).

6 Addendum for Additional Analyses

In the study, due to low enrollment in the first wave of internet platforms, it was rebooted for a second time following the end of the second wave, and the data from both the first and second iterations of the first wave were combined for statistical analysis of the first wave.

7 Appendix

Primary outcome table shell:

Type of platform	Wave	Number of days for each wave	Number of test kits ordered	Order rate (ordered test kits/day)
Social media site				
Facebook	1			
Instagram	2			
Subtotal				
Dating app				
Grindr	1			
Jack'D	2			
Subtotal				
Information search site				
Google	1			
Bing	2			
Subtotal				
Total				

Exploratory outcome table shell:

Platform	Impressions	Clicks	Click-through rate (%)	Users screened	Enrolled participants	Total funds spent (US \$)
Social media						
Dating apps						
Information search sites						

8 References

Protocol document for this study:

<https://datashare.nida.nih.gov/sites/default/files/studydocs/add/CTN0083-Protocol-Social-Media-PrEP.pdf>

Appendix that details the primary outcome model:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9591705/#app1>

Statistical Analysis Plan Checklist

Below you will find a checklist of recommended items to include in a statistical analysis plan. Some of these are specific to clinical trials (based on this [JAMA paper](#)) and some are other are specific to observational studies (based on [STROBE](#)/[RECORD](#) guidelines), so every item will not be necessary for every project. The biostatistician should start with the SAP template above and add in necessary information from the checklist. Item numbers that are starred (*) are not explicitly included in the SAP template and should be added by the author if relevant to the project. This checklist was developed using the [CONSORT 2010 Checklist](#).

Section/Topic	Item #	Description	Included (Yes/No/NA)
Administrative Information			
Study Information	1a	Descriptive title that matches the protocol, with SAP either as a forerunner or subtitle	_____
	1b	Trial registration number, protocol version number, and/or IRB number.	_____
	1c	CRU/Department/Division/Center/other collaborative unit that the study falls under	_____
Roles and responsibility	2a	Listing of principal investigators, clinical leads, and co-authors (if known)	_____
	2b	Name and affiliation of SAP author(s)	_____
	2c	Names, affiliations, and roles of other SAP contributors (e.g. senior statistician)	_____
SAP Information	3	SAP version number, with date of current version and original creation date	_____
Project Information	4a	Project folder location	_____
	4b	Project goals (e.g. manuscript, abstract, presentation, etc.)	_____
	4c	Project deadlines (of listed goals)	_____
	4d	Effort estimate	_____
Investigator Agreement			
Investigator Agreement	5	Confirmation that BERD Method Core's collaborative process has been reviewed, that all statistical analyses included in an abstract or manuscript should reflect the SAP, no changes should be made to the SAP without discussing with the SAP author, all biostatisticians on the SAP are co-authors on the manuscript, and that publications resulting from the SAP must cite grant number UL1TR002553 and be submitted to PubMed Central	_____
Signatures	6	Signatures of SAP author, senior statistician, and principal investigator(s)	_____
Activity Log			
SAP revisions	7a	SAP revision history with dates	_____
	7b	Justification for each SAP revision	_____
	7c*	Timing of SAP revision in relation to any interim analyses or submissions	_____
Study Overview			

Background and introduction	8	Synopsis of scientific background and rationale for the study	
Aims and Hypotheses	9a	List of all scientific aims/objectives of the study, with specifications of primary, secondary, etc.	
	9b	List of all statistical hypotheses (corresponding to the scientific aims), with specifications of primary, secondary, etc.	
Variables of Interest	10a	List of all outcome/endpoint variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	
	10b	List of all exposure variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	
	10c	List of any additional variables of interest (e.g. covariates, potential confounders, effect modifiers, etc.) in the analysis	
	10d*	Location of data dictionary (or provided as an appendix)	
	10e*	Report category boundaries if continuous variables are collapsed into categories, and describe any other relevant data transformations	
Causal Graph	11*	May be helpful to include a DAG or other graph/diagram that describes the way the variables of interest are presumed to relate to each other	
Study Methods			
Study Plan and Design	12a	Description of the study design (e.g. parallel group randomized trial, case-control study, cohort study, etc.)	
	12b*	Study setting, location, and relevant dates (e.g. periods of enrolment, exposure, follow-up, and collection)	
	12c*	Description of intervention or exposure groups, with allocation ratios, and details of any matching criteria	
	12d*	Details on randomization (e.g. stratification factors) and blinding procedures	
	12e	List of eligibility and/or inclusion/exclusion criteria	
	12f*	Description of screening/enrolment/recruitment processes	
	12g*	Description of patient flow (e.g. CONSORT diagram)	
	12h*	Description of analysis population (e.g. intention to treat, per protocol, etc.)	
	12i*	Definitions of adherence/compliance, protocol deviations, loss-to-follow-up, adverse events, etc.	
	12j*	Time points at which outcomes are measured	
	12k*	Timing of final analyses (are all outcomes analysed collectively, or will short-term outcomes be analysed separately from long-term outcomes, etc.)	

Sample Size	13a*	Sample size calculation or justification (either provided in full or summarized, with link to original source)	
	13b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
	14a*	Description of what interim analyses will be conducted at which time points, and what methods used to adjust significance levels due to the interim analysis	
	14b*	Details of any guidelines (e.g. safety, futility) for stopping the study early	
	14c*	Details of any changes to trial design due to interim analyses (e.g. enrolling more patients)	
Interim Analyses	15a	Description of data collection/acquisition process, with contact information for team member responsible	
	15b	Description of data flow/transfer from primary data collection through to creation of final analysis dataset	
	15c	Data transfer method and date	
	15d	Folder location where datasets are stored	
	15e*	Description of any additional data management, quality control, or processing undertaken	
	15f*	If any data are extracted from a database, a description of the database and the query used for the extraction, and whether/how it was merged with any data from outside that database. If the study involved linkage of databases, consider use of a flow diagram to demonstrate the data linkage process, including the number of individuals with linked data at each stage.	
	15f*	Description of any other data sources incorporated in the analysis	
Missing Data	16a*	Description of sources and magnitudes of missing data	
	16b*	Description of how missing data patterns will be presented/summarized (may be helpful to have a table shell or draft CONSORT-style diagram)	
	16c*	Description of contingency plans for handling missing data in analysis	
Simulations	17a*	If conducting a simulation, a description of the purpose of the simulation and its design (e.g. fully factorial, partially factorial, grid search, etc.)	
	17b*	Define the fixed and variable factors or parameters in the simulation, the estimands/targets of the simulation, and the performance measures to be estimated (with justifications of their relevance to the estimands/targets)	
	17c*	Description of the tabular and graphical presentations of simulation results and their interpretation	
Statistical Analysis Plan			
Statistical Significance	18a*	Hypothesis testing framework (e.g. superiority, equivalence, non-inferiority), or description of alternative analytic framework (e.g. evaluation of a posterior in a Bayesian analysis, etc.)	

Descriptive Statistics	18b*	Level of significance for primary hypotheses, including a description and rationale for any multiple comparisons adjustment or Type I error control procedures	
	18c*	Description of any decision-making rules based on confidence intervals, credible intervals, prediction intervals, Bayes' factors, or other alternative inferential methods	
	18d*	Description of how the results of any hypothesis tests (or alternative inferential methods) will be interpreted with respect to both the statistical hypotheses and scientific aims/objectives of the study	
	19a*	List of characteristics (e.g. demographic, clinical) to be summarized descriptively (e.g. "Table 1")	
	19b*	Description of how these characteristics will be summarized descriptively (e.g. means/medians vs. N (%), tabular displays, graphical displays, etc.)	
	19c*	Summarize follow-up time (e.g. average and total amount) and number of events	
Analysis Methods	20a	For each aim/hypothesis (see items 9a/9b), a description of what analysis method will be used and how the results from this method will be reported and interpreted	
	20b*	Description of any transformations, standardizations, covariate or confounder adjustments, weighting, or stratification methods to be used and why.	
	20c*	For each analytic method proposed, a description of the assumptions of that method and what processes will be used to evaluate whether or not those assumptions hold	
	20d*	Details of contingency plans/alternative methods to be used if the assumptions are found not to hold	
	20e*	In the case of non-standard test statistics, formulas provided for the test statistic with a description of the mathematical null hypothesis, how significance is determined, and how the test statistic is interpreted	
	20f*	In the case of regression models, formulas provided for the full model with a description of which parameters are to be used, how they will be interpreted, how confidence intervals will be constructed, etc.	
	20g*	In the case of survey, hierarchical/nested, or clustered data, a description of what methods will be used to adjust for the data structure and why (e.g. if using a GEE, describing which correlation structure and why it was chosen, etc.)	
	20h*	For non-continuous outcomes, clearly explain the effect used (e.g. risk difference, risk ratio, odds ratio, etc.), whether it is relative or absolute, and justify why that was chosen as the effect measure of interest	
	20i*	Documentation of any non-standard methods used (e.g. using alternative degree of freedom calculation methods, using a non-canonical link function, etc.)	
	20j*	Description of any limitations, sources of bias, internal/external validity, and other relevant discussions concerning the interpretation and generalizability of the design or methods used	
Additional Analysis Methods	21a*	Description of any pre-planned sensitivity analyses and how they will be interpreted	
	21b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
	21c*	Description of any additional post-hoc calculations or analyses (e.g. evaluating interaction/modification effects, calculating mediation or local average treatment effects, evaluation of AUROC curves, etc.)	

Exploratory Analyses	21d*	If conducting any bootstrap analyses, a description of the sampling algorithm and number of iterations used	
	21e*	If conducting any cross-validation procedures, a description of how the cross-validation is conducted (e.g. leave-one-out, train/validation/test, etc.)	
	22a*	Description and justification for any pre-planned exploratory analyses and what methods will be used to conduct them	
	22b*	Framework for conducting any unplanned exploratory analyses and how they will be integrated into the planned analysis	
Software	23*	List of statistical software (along with version numbers) to be used for each phase of the analysis; in the case of R or Stata, additionally list any requisite installed packages and their version numbers	
Other	24*	Description of any additional planned analyses of the data (e.g. a safety analysis looking at adverse event rates for a Data Safety Monitoring Board, etc.)	
Tables and Figures			
Table Shells	25*	Example tables related to any of the conducted analyses; if possible including any available preliminary data	
Example Figures	26*	Example figures related to any of the conducted analyses; if possible including any available preliminary data.	
References			
References	27a	References for any non-standard statistical methods used	
	27b	References (and locations) for any relevant protocols, standard operating procedures, or other documents cited in the SAP	
Additional Information			
Appendices	28*	If necessary, appendices may be included (e.g. a full data dictionary, a copy of a Case Report Form, etc.)	
Addendums	29*	Any additional analyses conducted that were not included in the SAP should be documented in an addendum, describing the purpose of the additional analysis, when it was conducted, and by whom	